



Aplastic Anemia

(Hypoplastic Anemia)

By [Gloria F. Gerber](#), MD, Johns Hopkins School of Medicine, Division of Hematology
Reviewed/Revised Apr 2026

Aplastic anemia is a disorder in which the cells of the bone marrow that develop into mature blood cells are destroyed, leading to low numbers of red blood cells, white blood cells, and/or platelets.

When the bone marrow cells (stem cells) that develop into mature blood cells and platelets are damaged or destroyed, the bone marrow can shut down. This bone marrow failure is called aplastic anemia. The bone marrow failure leads to too few red blood cells ([anemia](#)), too few white blood cells ([leukopenia](#)), and too few platelets ([thrombocytopenia](#)).

The term aplastic anemia is used to refer to anemia that occurs when production of most or all types of blood cells is suppressed. If production of only red blood cells is suppressed, the disorder is called pure red blood cell aplasia.

When the cause of aplastic anemia cannot be diagnosed (called idiopathic aplastic anemia), the cause is likely an [autoimmune disorder](#), in which the immune system suppresses bone marrow stem cells.

Other causes include:

- Infection with viruses such as parvovirus (causes only pure red blood cell aplasia), [Epstein Barr virus](#), or [cytomegalovirus](#)
- Radiation exposure
- Toxins (such as benzene)
- Chemotherapy agents and other medications (such as chloramphenicol)
- Pregnancy
- [Hepatitis](#)

Symptoms of Aplastic Anemia

Symptoms of aplastic anemia usually develop slowly over weeks to months.

The anemia causes fatigue, weakness, and paleness. The leukopenia causes increased susceptibility to infection. The thrombocytopenia causes easy bruising and bleeding.

Diagnosis of Aplastic Anemia

- Blood tests
- Bone marrow examination

Blood tests are done in people who have symptoms of anemia. When blood tests show that all blood cell numbers are reduced, a bone marrow examination is done.

Bone Marrow Biopsy and Bone Marrow Aspiration



Aplastic anemia is diagnosed when microscopic [examination of a sample of bone marrow](#) (bone marrow biopsy) reveals a sharp decrease in bone marrow cells.

Treatment of Aplastic Anemia

- Transfusions
- Stem cell transplantation

People with severe aplastic anemia may die unless immediately treated. [Transfusions](#) of red blood cells, platelets, and substances called growth factors may temporarily increase the numbers of red blood cells, white blood cells, and platelets.

Stem cell transplantation is a treatment for aplastic anemia and cures the disease. Alternatively, people are given antithymocyte globulin and [cyclosporine](#) to suppress the immune system and allow the bone marrow stem cells to regenerate. [Eltrombopag](#), which stimulates the bone marrow cells, is sometimes given with these medications.

Drug Information for the Topic
